Combined Architecture: Linux + Cloud + DevSecOps

This unified architecture diagram represents the integration of three core layers:

1. **Linux Server Administration Layer**:

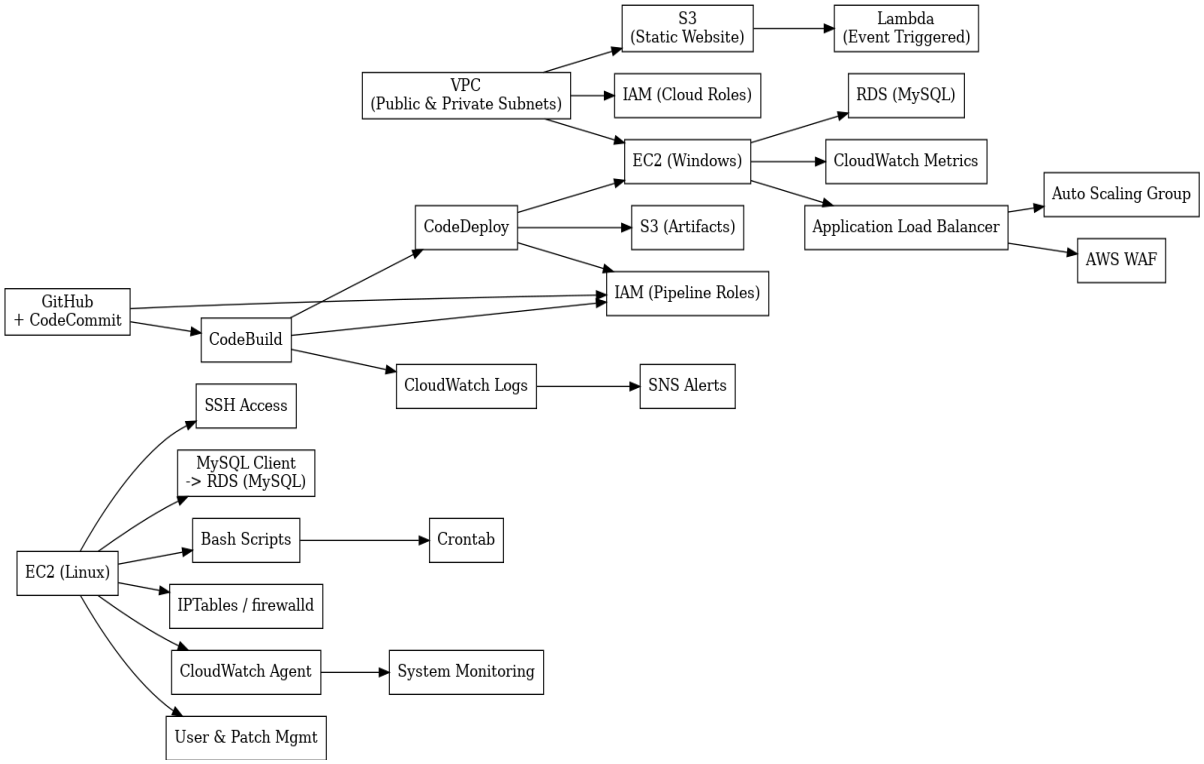
- AWS EC2 instances run Linux with SSH access.
- Automated operations using Bash scripting and Crontab.
- IPTables and firewalld handle custom firewall configurations.
- System metrics collected via CloudWatch Agent with host-level monitoring.
- User management and patching follow best security practices.

2. **Cloud Infrastructure Layer**:

- A custom VPC with both public and private subnets supports isolated environments.
- EC2 instances (Windows) host backend services.
- Load balancing via ALB and scalability via Auto Scaling Groups.
- Static website content hosted on S3 and RDS (MySQL) used for database operations.
- IAM enforces security policies, CloudWatch provides visibility, and WAF ensures protection.
- Lambda is triggered by events like file uploads.

3. **DevSecOps CI/CD Pipeline Layer**:

- CodePipeline integrates GitHub, CodeCommit, CodeBuild, and CodeDeploy.
- Build artifacts are deployed to both S3 and EC2.
- Monitoring and alerts are managed via CloudWatch Logs and SNS.
- IAM policies ensure security throughout the CI/CD process.


S3
(Static Website)
Lambda
(Event Triggered)
VPC
(Public & Private Subnets)
IAM (Cloud Roles)
RDS (MySQL)
EC2 (Windows)
CloudWatch Metrics
Auto Scaling Group
CodeDeploy
S3 (Artifacts)
Application Load Balancer
AWS WAF
IAM (Pipeline Roles)
GitHub
+ CodeCommit
CodeBuild
CloudWatch Logs
SNS Alerts
SSH Access
MySQL Client
-> RDS (MySQL)
Bash Scripts
Crontab
EC2 (Linux)
IPTables / firewalld
CloudWatch Agent
System Monitoring
User & Patch Mgmt